Global Coffee & Living Index: Econometric Analysis, Statistical Modeling, and Data Pipeline Quality Report

A Comprehensive Cross-Sectional Analysis of Real Purchasing Power, Price Elasticities, and Agricultural Commodity Traps

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Data Vintage: World Bank Indicators (2018 - 2023) & FAOSTAT Coffee Statistics (2020 - 2022)

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Abstract

This report presents an econometric investigation and data quality evaluation of the *Global Coffee & Living Index* ($n = 251$ world entities, $n = 198$ fully matched observations). We examine global income distribution, domestic price elasticity, real purchasing power dynamics, and agricultural trade imbalances. Log-log Ordinary Least Squares (OLS) regression confirms the existence of the Penn Effect ($\epsilon = 0.2252, R^2 = 0.572$), establishing that domestic price levels increase at less than one-quarter the rate of national income growth. Consequently, real domestic purchasing power accelerates non-linearly in developed economies ($R^2 = 0.777$).

Furthermore, we provide empirical evidence of the *Coffee Commodity Paradox*: green coffee-producing nations exhibit an average Living Affordability Score (LAS) of 9.35 compared to 29.29 for non-producing entities ($p < 10^{-13}$). Multivariate OLS models confirm that green coffee production volume yields no statistically significant positive contribution to domestic living standards ($\beta = 0.1812, p = 0.348$). Finally, we document the remediation of silent API pipeline failures and propose an architectural roadmap for publication.

Contents

1. Executive Summary & Research Objectives

The *Global Coffee & Living Index* synthesizes macroeconomic indicators from the World Bank API, green coffee production metrics from the UN Food and Agriculture Organization (FAOSTAT) API, and geographic capital metadata. The primary objective is to measure domestic purchasing power across nations while assessing whether participation in primary agricultural commodity production (specifically coffee) translates into tangible local living standards.

1.1. Key Analytical Findings

1. **Non-Linear Acceleration of Purchasing Power:** While the Price Level Index (PLI) scales positively with Gross National Income (GNI), its income elasticity is inelastic ($\epsilon = 0.2252$). As a result, living affordability (LAS) accelerates rapidly in upper-income brackets, resulting in high global inequality ($\text{Gini}_{LAS} = 0.5281$).

2. **The Agricultural Commodity Paradox:** Major coffee-producing nations are concentrated in lower-income brackets. Controlling for income, coffee production volume demonstrates no statistically significant relationship with local living standards ($p = 0.348$), highlighting an ongoing value-retention bottleneck within the global agricultural supply chain.

3. **Pipeline Defect Remediation:** Audit procedures resolved silent API timeout failures, preventing 100% missingness in agricultural fields and eliminating 48 non-country aggregate regions from leaking into country-level cross-sections.

2. Methodological Framework & Metric Formulation

To quantify local living standards relative to macroeconomic output, the dataset constructs two continuous derived metrics. Let GNI_i denote Gross National Income per capita (Atlas method, current US$) for country i, and let PLI_i denote the Price Level Index (expressed relative to the United States baseline of 100).

2.1. Purchasing Power Ratio (PPR)

The Purchasing Power Ratio (PPR_i) measures real local income per unit of domestic price intensity:

$$PPR_i = \frac{GNI_i}{PLI_i} \tag{1}$$

Because PLI_i reflects the cost of a standardized basket of goods relative to the U.S., PPR_i acts as a price-adjusted nominal income metric.

2.2. Living Affordability Score (LAS)

To project PPR_i onto a bounded continuous scale $[0, 100]$ suitable for cross-country benchmarking, min-max feature normalization is applied:

$$LAS_i = \left(\frac{PPR_i - \min(PPR)}{\max(PPR) - \min(PPR)} \right) \times 100 \tag{2}$$

Across the matched baseline sample ($n = 198$), observed boundary values are $\min(PPR) = 12.86$ (Burundi) and $\max(PPR) = 1,277.93$ (Qatar).

3. Parametric Summary & Statistical Distributions

Table 1 summarizes central tendency and dispersion metrics across the full cross-sectional dataset ($n = 251$).

Indicator Variable	n	**Mean**	**Std Dev**	**Min**	**Median**	**Max**
GNI per capita (US$)	247	$17,631.56	$23,011.40	$280.00	$7,210.00	$133,460.00
Price Level Index (PLI)	202	50.33	22.36	18.64	43.17	117.05
PPP Conversion Factor	202	850.39	6,888.66	0.17	5.32	94,280.27
Purchasing Power Ratio (PPR)	198	291.63	281.93	12.86	190.12	1,277.93
Living Affordability Score (LAS)	198	22.04	22.29	0.00	14.01	100.00
Coffee Production Qty (t)	77	144,100.90	439,259.90	1.54	8,557.00	3,179,341.00

Table 1: Parametric summary statistics across macroeconomic and agricultural variables.

3.1. Percentile Distribution & Higher-Order Moments

Both national income and living affordability exhibit substantial positive skewness and heavy right tails. Table 2 details distributional percentiles and higher-order moments.

Percentile / Moment	**GNI per capita**	**PLI**	**PPP Factor**	PPR	LAS
5th Percentile (P_5)	$770.00	24.56	0.43	20.00	0.57
10th Percentile (P_{10})	$1,220.00	27.66	0.54	33.67	1.64
25th Percentile (P_{25})	$2,570.00	32.77	0.94	73.18	4.77
50th Percentile (P_{50})	$7,210.00	43.17	5.32	190.12	14.01
75th Percentile (P_{75})	$22,085.00	63.21	93.33	429.44	32.93
90th Percentile (P_{90})	$54,050.66	85.00	434.43	730.28	56.71
95th Percentile (P_{95})	$71,315.00	92.11	1,338.77	810.88	63.08
99th Percentile (P_{99})	$93,707.20	111.18	10,392.53	1,166.17	91.17
Skewness (γ_1)	**1.992**	**0.917**	**12.737**	**1.328**	**1.328**
Excess Kurtosis (γ_2)	**4.060**	**0.058**	**171.050**	**1.229**	**1.229**

Table 2: Percentile break points, skewness, and excess kurtosis coefficients ($n = 198$).

The GNI skewness ($\gamma_1 = 1.992$) underscores global wealth concentration: the mean GNI ($17,631.56) exceeds the 50th percentile ($7,210.00) by a factor of 2.44.

4. Econometric Modeling & Empirical Analysis

4.1. Price-Income Elasticity: The Penn Effect

Economic theory predicts that domestic price levels rise with per-capita income due to non-tradable service productivity differentials (the Balassa-Samuelson effect and Penn Effect). To estimate global price elasticity, we evaluate a log-log OLS regression model:

$$\ln(PLI_i) = \beta_0 + \epsilon \cdot \ln(GNI_i) + u_i \tag{3}$$

Variable	Coefficient	Std. Error	t-statistic	p-value	95% Conf. Interval
Constant (β_0)	1.8034	0.126	14.297	< 0.0001	$[1.555, 2.052]$
$\ln(GNI)$ (ϵ)	0.2252	0.014	16.194	< 0.0001	$[0.198, 0.253]$
Model Diagnostics: $n = 198$, $R^2 = 0.572$, Adjusted $R^2 = 0.570$, $F(1, 196) = 262.3, p < 10^{-37}$					

Table 3: Log-log OLS estimation of Price Level Index elasticity relative to GNI per capita.

Econometric Interpretation: The estimated elasticity $\hat{\epsilon} = 0.2252$ ($p < 0.0001$) confirms that a 10% increase in GNI per capita is associated with only a 2.25% increase in domestic price levels. Because domestic prices increase at less than one-quarter the rate of nominal wage growth, high-income nations experience substantial real gains in domestic purchasing power.

4.2. Living Affordability Dynamics

To evaluate how real purchasing power accelerates with national wealth, we estimate a semi-log model predicting LAS_i directly from $\ln(GNI_i)$:

$$LAS_i = \alpha_0 + \alpha_1 \cdot \ln(GNI_i) + e_i \tag{4}$$

Variable	Coefficient	Std. Error	t-statistic	p-value	95% Conf. Interval
Constant (α_0)	-101.2896	4.777	-21.203	< 0.0001	$[-110.711, -91.868]$
$\ln(GNI)$ (α_1)	13.7662	0.527	26.140	< 0.0001	$[12.728, 14.805]$
Model Diagnostics: $n = 198$, $R^2 = 0.777$, Adjusted $R^2 = 0.776$, $F(1, 196) = 683.3, p < 10^{-65}$					

Table 4: Semi-log OLS estimation of Living Affordability Score relative to log GNI per capita.

The model demonstrates strong explanatory power ($R^2 = 0.777$). Each 1-unit increase in $\ln(GNI)$ yields a 13.77 point increase in LAS.

5. Global Inequality & Income Quartile Analysis

To measure inequality across macroeconomic metrics, relative concentration indices were calculated. The dataset exhibits a Gini coefficient of 0.6155 for GNI per capita, 0.5048 for PPR, and 0.5281 for LAS.

Table 5 divides the matched sample ($n = 198$) into four income quartiles.

Income Quartile	Count	Mean GNI	Mean PLI	Mean PPR	Mean LAS	Coffee Producers
Q1 (Low Income)	50	$1,389.60	32.45	44.78	2.52	33 (66.0%)
Q2 (Lower-Middle)	49	$4,888.57	40.93	131.71	9.39	22 (44.9%)
Q3 (Upper-Middle)	49	$14,029.80	50.36	290.88	21.98	15 (30.6%)
Q4 (High Income)	50	$53,058.60	76.16	695.95	54.00	2 (4.0%)

Table 5: Disaggregated structural breakdown across GNI per capita quartiles.

Quartile Observations:

- Affordability Gap: Q4 entities enjoy a mean LAS 21.4 times higher than Q1 entities (54.00 vs. 2.52), despite Q4 price levels being only 2.3 times higher (76.16 vs. 32.45).

- Agricultural Clustering: 66.0% of Q1 nations produce green coffee, compared to 4.0% of Q4 nations.

6. The Coffee Commodity Paradox

A central question is whether national coffee production supports domestic purchasing power.

6.1. Hypothesis Testing: Producers vs. Non-Producers

We evaluate the difference in Living Affordability Scores between coffee-producing nations ($n = 72$) and non-producing entities ($n = 126$).

Group Cohort	**Sample (n)**	**Mean LAS**	**Median LAS**	**Mean GNI per capita**
Coffee-Producing Nations	72	9.35	6.21	$4,812.50
Non-Producing Entities	126	29.29	23.34	$24,952.18
Disparity Ratio	-	**3.13**×	**3.76**×	**5.18**×

Table 6: Comparison of living standards between coffee-producing and non-producing entities.

Statistical Significance Tests:

- Welch's t-test (Unequal Variances): $t = -8.1241, p = 6.42 \times 10^{-14}$.
- Mann-Whitney U Non-Parametric Test: $U = 2048.0, p = 1.42 \times 10^{-10}$.

Both tests reject the null hypothesis, confirming that coffee-producing nations experience significantly lower living affordability scores than non-producers.

6.2. Multivariate OLS Regression Within Coffee Producers

To determine if production volume provides a marginal benefit, we estimate a multivariate OLS model for coffee producers ($n = 72$):

$$LAS_i = \gamma_0 + \gamma_1 \cdot \ln(GNI_i) + \gamma_2 \cdot \ln(\text{Coffee Qty}_i) + \eta_i \tag{5}$$

Variable	**Coefficient**	**Std. Error**	**t-statistic**	**p-value**	**95% Conf. Interval**
Constant (γ_0)	-58.1037	5.248	-11.071	< 0.0001	$[-68.574, -47.633]$
$\ln(GNI)$ (γ_1)	8.0999	0.584	13.875	< 0.0001	$[6.935, 9.264]$
ln(Coffee Qty) (γ_2)	0.1812	0.192	0.944	**0.3480**	$[-0.202, 0.564]$
Model Diagnostics: $n = 72$, $R^2 = 0.737$, Adjusted $R^2 = 0.729$, $F(2, 69) = 96.43, p < 10^{-20}$					

Table 7: Multivariate OLS model assessing marginal impact of production volume on LAS.

Analytical Conclusion: Controlling for national income, the coefficient for coffee production volume ($\gamma_2 = 0.1812, p = 0.3480$) is not statistically significant. Increasing coffee output volume does not independently improve local purchasing power, illustrating a primary agricultural commodity trap where retail value captures occur downstream in consuming markets.

7. Cross-Country Comparative Outlier Analysis

Table 8 highlights the top 10 and bottom 10 performing nations in Living Affordability Scores.

Top 10 Living Affordability Scores					Bottom 10 Living Affordability Scores				
ISO	**Country**	**GNI ($)**	**PLI**	**LAS**	**ISO**	**Country**	**GNI ($)**	**PLI**	**LAS**
QAT	Qatar	79,430	62.16	100.00	BDI	Burundi	280	21.77	0.00
NOR	Norway	101,840	84.43	94.34	CAF	Cent. Afr. Rep.	530	39.42	0.05
SGP	Singapore	69,530	59.69	91.07	MOZ	Mozambique	540	37.03	0.14
BMU	Bermuda	133,460	117.05	89.11	SOM	Somalia	600	38.34	0.22
MAC	Macao SAR	64,780	57.73	87.68	LBR	Liberia	710	44.50	0.24
LUX	Luxembourg	88,480	87.31	79.09	COD	Dem. Rep. Congo	680	38.12	0.39
IRL	Ireland	79,280	80.05	77.27	MDG	Madagascar	510	28.40	0.40
BRN	Brunei	34,390	38.65	69.32	MWI	Malawi	620	34.56	0.40
CHE	Switzerland	98,160	111.19	68.77	NER	Niger	640	34.05	0.47
DNK	Denmark	72,080	87.88	63.82	AFG	Afghanistan	370	18.78	0.54

Table 8: Cross-country rankings of top 10 and bottom 10 entities by Living Affordability Score.

7.1. Case Studies

- Qatar ($LAS = 100.00$): Qatar ranks first due to high per-capita GNI ($79,430) combined with a moderate price index ($PLI = 62.16$), driven by energy subsidies and low import tariffs.
- Burundi ($LAS = 0.00$): Burundi exhibits the global minimum PPR (12.86). Despite low prices ($PLI = 21.77$), extremely low nominal GNI ($280) restricts real purchasing power.

8. Data Pipeline Audit & Completeness Analysis

Systematic code auditing identified five pipeline software defects, detailed in Table 9.

8.1. Completeness Matrix

Table 10 outlines missingness patterns across all 251 output records.

9. Strategic Recommendations & Architectural Roadmap

9.1. Data Pipeline Engineering

1. Direct API Pipeline Execution: Execute scripts `01–04` with live internet access to query the FAOSTAT API directly and populate export value metrics (`coffee_export_value_usd1000`).
2. Log Transformations for Modeling: Include a $\log(PPR)$ variable in the release file to mitigate positive skewness ($\gamma_1 = 1.328$) for downstream regression tasks.
3. Dynamic Region Filtering: Replace static ISO exclusion arrays with dynamic API structural metadata checks to auto-detect aggregate region flags.

Severity	Pipeline Defect Description	Engineering Resolution
Critical	Script `02` failed silently during API timeouts, outputting an empty CSV and causing 100% missingness in coffee metrics.	Added multi-year API retry loops (2022 → 2021 → 2020), offline CSV fallback parsers, and validation guards.
Critical	World Bank regional aggregates (e.g., `WLD`, `EUU`) leaked into country datasets because ISO code length checks (`len == 3`) failed to filter aggregate entities.	Replaced string length checks with a hardcoded exclusion list of 48 regional aggregate ISO codes.
Moderate	Coffee export value columns were missing when executing via manual bulk fallback files.	Documented fallback behavior in execution logs and implemented non-null coverage checks.
Moderate	Observation year columns were discarded during extraction, reducing auditability.	Preserved source metadata fields (`gni_year`, `pli_year`, `coffee_data_year`).
Moderate	Dataset release year (2026) was confused with source data vintage (2018–2023).	Standardized documentation to distinguish publication release year from underlying survey periods.

Table 9: Pipeline error audit, severity classification, and applied resolutions.

9.2. Economic & Policy Insights

1. Domestic Value Retention: Policy initiatives in coffee-producing nations should focus on roasting, processing, and direct-to-consumer exports rather than raw bean production.

2. Targeted Affordability Benchmarking: Organizations should use PPR alongside CPI indicators to evaluate real wage purchasing power in developing economies.

10. Conclusion

The *Global Coffee & Living Index* offers an empirical framework for evaluating purchasing power and agricultural economics across $n = 198$ matched nations. Econometric analysis confirms that domestic price levels scale inelastically relative to national wealth ($\epsilon = 0.2252$), driving rapid gains in real living affordability in developed economies.

Furthermore, hypothesis testing and multivariate regression reveal that green coffee production volume does not yield a statistically significant improvement in domestic purchasing power ($p = 0.348$). Resolving pipeline data quality issues ensures that the public release dataset provides a clean foundation for econometric research.

Field Variable Name	Data Type	Non-Null Count	Missingness (%)
country_name	String	247 / 251	1.6%
iso3_code	String	251 / 251	0.0%
gni_per_capita_usd	Float64	247 / 251	1.6%
price_level_index	Float64	202 / 251	19.5%
ppp_conversion_factor	Float64	202 / 251	19.5%
representative_city	String	204 / 251	18.7%
coffee_production_qty	Float64	77 / 251	69.3%
production_unit	String	80 / 251	68.1%
coffee_export_value_usd1000	Float64	0 / 251	100.0%
export_unit	Float64	0 / 251	100.0%
purchasing_power_ratio	Float64	198 / 251	21.1%
living_affordability_score	Float64	198 / 251	21.1%

Table 10: Completeness matrix for the consolidated dataset ($n = 251$).

Appendix: Data Dictionary & Variable Documentation

Field Name	Operational Definition & Unit of Measure
country_name	Official entity name standardized from World Bank metadata.
iso3_code	ISO 3166-1 alpha-3 international country identifier (Primary Key).
gni_per_capita_usd	Gross National Income per capita (Atlas method, current US$).
price_level_index	Price Level Index relative to United States baseline ($US = 100$).
ppp_conversion_factor	Purchasing Power Parity conversion factor (Local Currency Units per Intl $).
representative_city	Designated capital city name for spatial mapping.
coffee_production_qty	Volume of green coffee produced annually (Metric tonnes, t).
production_unit	Unit specification for agricultural output (Standardized to 't').
coffee_export_value_usd1000	Annual export value of green coffee ($1,000 USD).
export_unit	Unit specification for export values (Standardized to '1000 US$$Short$').
purchasing_power_ratio	Derived metric representing per-capita income per price unit (GNI/PLI).
living_affordability_score	Min-max normalized purchasing power score projected onto $[0, 100]$.